

Organ dose measurements using an adult anthropomorphic phantom and risk estimation of cancer incidence from CBCT exposures

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ABSTRACT

Cone-beam CT (CBCT) has become an essential tool for pre-treatment verification of the patient's position and for targeting the tumor volume localization in Image Guided Radiotherapy (IGRT). CBCT imaging is employed generally on a daily-basis, for each treatment fraction, and several times per patient, to ensure the correct patient's set-up. This leads to cumulative imaging doses to the tissues surrounding the exposed target-organs. The objective of this work is to determine the patient organ doses from a thorax CBCT scan in order to estimate risk of cancer incidence due to CBCT exposures.

A thorax CBCT scan was performed in an anthropomorphic phantom of an adult male (CIRS ATOM) and the organ doses were assessed from point measurements using Thermo Luminescent Detectors (TLDs). The measurements were performed using a CBCT imaging system mounted on a LINAC (EdgeTM, Varian Medical Systems). The lifetime attributable risk (LAR) of cancer incidence was determined using the BEIR risk models. Considering a single thorax CBCT scan, the highest organ doses were calculated for heart and left lung which registered values of 5.84 ± 0.99 mGy and 4.90 ± 0.83 mGy, respectively. In contrast, the lowest organ dose was determined for right lung, with an absorbed dose of 3.07 ± 0.52 mGy. Regarding risk estimation, after a complete course of IGRT treatment for lung cancer (24 fractions) and assuming that, at least, one CBCT scan is performed per fraction, the LAR of cancer incidence varies between 27 to 309 cases per 100.000 exposed persons, depending on the organ evaluated. This work highlights the need to determine radiation induced cancer risks arising from CBCT repeated exposures in order to optimize the selected scanning protocols and consequently the radiological protection of patient. Furthermore, accurate organ dose calculation is fundamental to reduce the uncertainties associated with radiological risk estimation in imaging procedures.

1. Introduction

The effect of ionizing radiation (IR) on biological human tissues is, among others, a complex function of the absorbed dose. In the range of high doses (approximately from 1 Gy to 10 Gy), it is well established that radiation cause adverse health effects in humans such as acute radiation syndrome and acute death (Radiation Effects Research Foundation, 2019). In contrast, the effects of low doses (up to 100 mGy), such as those resulting from medical imaging exposures, may cause late effects, such as cancer, mainly due to -DNA mutations induced by radiation exposure in living cells (Radiation Effects Research Foundation, 2019). However, the exact mechanisms of these

mutations that could lead to cancer development are still not clear and considerable uncertainties remain.

In the low dose area, the exposure of individuals to IR due to medical imaging procedures is a cause of concern internationally, due to the significant contribution of the associated doses to the increase of the total population dose. The National Council on Radiation Protection and Measurements (NCRP) reported in 2009 the changes of IR exposure of the population of the United States (U.S.) between the 1980s and 2006. The estimated total radiation exposure per inhabitant in the U.S. was 3.6 mSv in the 1980s and increased to 6.2 mSv in 2006. The medical exposure registered an increase from 15% of the total exposure in the 1980s to 48% of the total exposure in 2006 (National Council on

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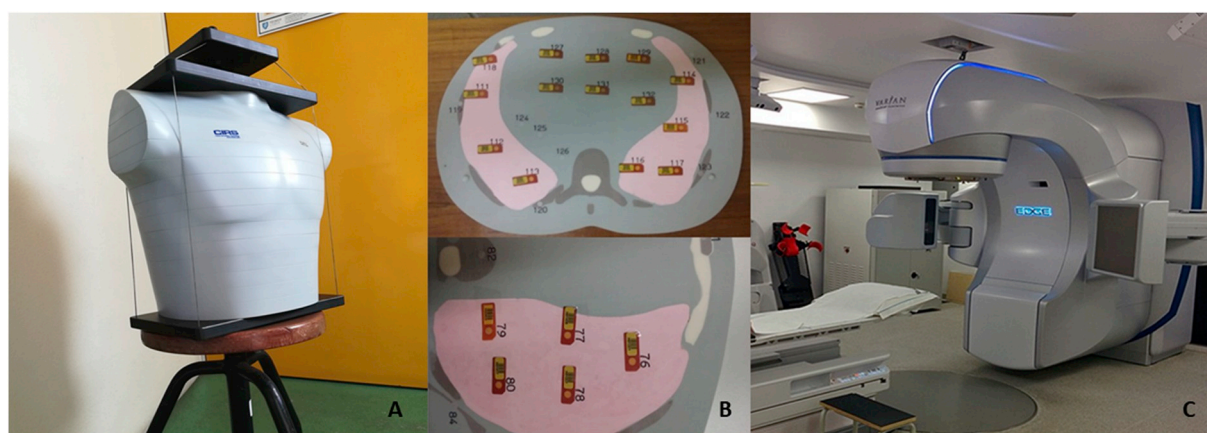


Fig. 1. A) 3D view of the thorax phantom used for measurements; B) example of some slice of the anthropomorphic CIRS phantom used where TLDs 100H were placed for dose measurements; C) OBI CBCT imaging system used for measurements, mounted on the gantry of the Varian Edge™ (Varian Medical Systems, Palo Alto, CA) LINAC at 90° to the therapeutic beam.

Radiation Protection and Measurements (NCRP, 2009).

Among the photon imaging techniques that nowadays are used for several purposes (i.e. diagnosis, screening, tumor patient localization, etc.), Computed Tomography (CT), is the one that more contributes to the increase of the mean effective dose per inhabitant (Kalender, 2014; International Commission on Radiological Protection (ICRP), 2007). One of the main technological advances in the CT X-ray field was the introduction of the Cone Beam CT (CBCT). This technique enables high-resolution volumetric scanning of the organs and soft tissues anatomy under investigation (American Association of Physicists in Medicine (AAPM), 2010; International Atomic Energy Agency (IAEA), 2011). An X-ray beam in the form of a rotating divergent cone or pyramid is measured by a Flat Panel detector. Acquisition of 2D-projection images are then reconstructed into a 3D-dataset (Sykes et al., 2013). In current clinical practice, CBCT scanners are being widely used for a variety of medical imaging applications ranging from radiotherapy, orthopedics, urology, dental-maxillofacial, neuro-interventions, vascular and non-vascular interventions (Rehaniet al., 2015). Their use is mainly due to the following characteristics: combining dynamic fluoroscopy and tomographic imaging; large z-axis range coverage of the patient's body anatomy and high-resolution imaging of high-contrast structures (Rehaniet al., 2015).

A very successful application of this imaging modality is in pre-treatment stage of Image Guided Radiotherapy (IGRT). According to a national survey performed by the American Society for Radiation Oncology, the use of CBCT in IGRT has grown from approximately zero in early 2000s to 92% in 2015 (Nabavizadehet al., 2016). The goal of radiotherapy treatment is to deliver the prescribed therapeutic radiation dose to the planned target volumes (PTV) while minimizing dose to critical normal organs at risk (OAR). For this reason, in order to minimize target position errors and improving accuracy and efficacy of radiotherapy plan, on board kV-CBCT modality is one of the most used imaging techniques in IGRT (ACR–AAPM, 2009; National Cancer Action Team, 2012). While the dose from a single CBCT scan is much lower than a fraction of therapeutic dose, CBCT imaging is used on a daily-basis, for each treatment fraction, and eventually several times per patient during any fraction, in order to ensure that the patient's position is correct, and to reposition if necessary. This may leads to high cumulative imaging doses to the healthy tissues surrounding the exposed target-organs (imaging volume much larger than treatment volume) (Rehaniet al., 2015; Ding et al., 2018). For this reason, CBCT dose imaging could be a non-negligible source of radiation dose to the patients' healthy tissues and the cumulative dose resulting from the fractionated treatments should be accurately assessed in order to minimize its risk (Rehaniet al., 2015; Baptista et al., 2019; Abuhaimed

et al., 2014). This concern is largely shared in several studies present in literature (Alaei and Spezi, 2015; Amer et al., 2007) and more recently, AAPM Task Group 180 report (Ding et al., 2018) considers 5% of the therapeutic target dose to be the threshold above which concomitant imaging dose should be accounted for in the treatment planning process. This choice is based on the analysis of clinical data that suggests that dose variations as small as 5% may lead to real variations in both tumor response and the risk of morbidity. Given the dose concerns above described, the aims of this study were, i) to assess the kV-imaging dose for a routine thorax protocol with an anthropomorphic physical phantom and Thermo Luminescent Detectors (TLDs) in target and surrounding healthy organs, ii) evaluate which could be the average associated cancer risk for patients undergoing these type of imaging exposure, both for a single kV-CBCT exposure and for the cumulative dose of an entire treatment.

2. Materials and methods

The measurements were made at Champalimaud Center for the Unknown (Lisbon, Portugal), with the On Board Imager (OBI) CBCT imaging system, mounted on the gantry of the Varian Edge™ (Varian Medical Systems, Palo Alto, CA) LINAC at 90° to the therapeutic beam, as shown in Fig. 1. All the exposure parameters of the thorax CBCT scanning protocol selected in this work are listed in Table 1. For the present study, the thorax scanning protocol was used with a half bowtie filter, allowing large FOVs by imaging body regions asymmetrically. Blades X1 and X2 were opened by + 2.4 cm and −24.7 cm at the isocentre, whereas Y1 and Y2 were positioned at ± 10.7 cm, at the isocentre (Baptista et al., 2019). The polychromatic X-ray spectrum of the OBI system is generated, according to the manufacture technical report, by a tungsten anode, 2.7 mm of thickness for the aluminum of inherent filter and 0.89 mm of titanium for beam hardening purposes (absorbing low photon energies) (True Beam Reference Guide, 2013). The mean energy for this spectrum is 65 keV and its energy distribution,

Table 1

Acquisition parameters of the Thorax CBCT scanning protocol used in this work.

Parameters	Thorax Scan Protocol
Tube Voltage (kV _p)	125
Tube current time product (mAs)	270
Gantry Rotation (°)	360
Acquisition mode	Half-fan
Beam width (mm)	214
Collimator Blades	X1 and X2 (cm) Y1 and Y2 (cm)
	−24.7 and + 3.4 −10.7 and + 10.7

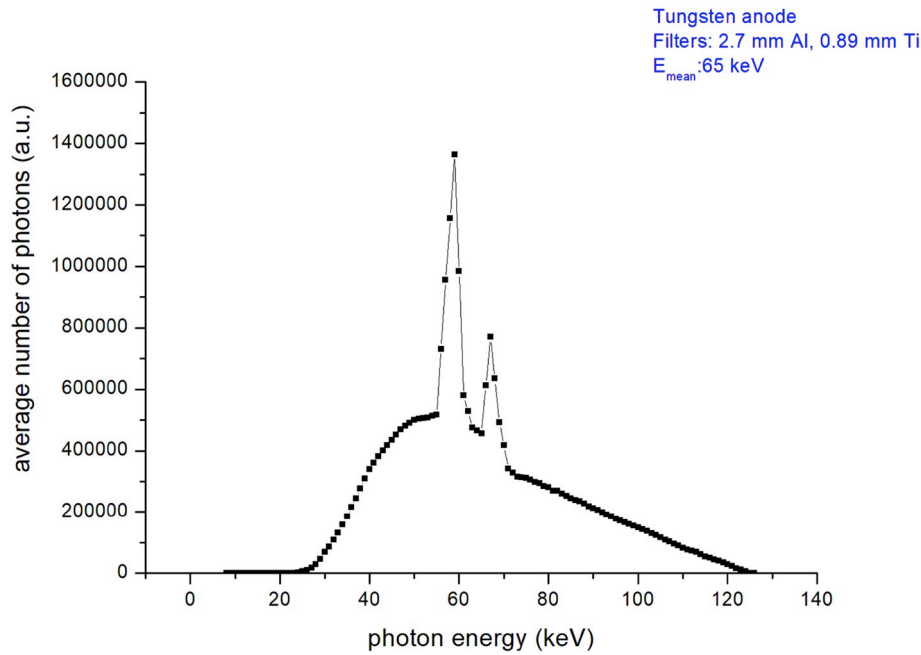


Fig. 2. X-ray energy spectrum obtained according to the specifications of the OBI kV imaging system.

generated by the semi-empirical tool developed by Boone et al. (Boone and Seibert, 1997), is showed in Fig. 2. A thorough work of quality check of the clinical kV beam used for measurements (i.e. beam profile, half-value layer assessment, etc.), also with Monte Carlo simulations, was performed in a previous work (Baptista et al., 2019).

2.1. Anthropomorphic phantom

An ATOM anthropomorphic phantom was used in this study for organ dose assessment [20.]. ATOM phantoms are constructed of CIRS proprietary tissue equivalent materials and are widely used for image quality and dose calibration. Linear attenuations of the simulated tissues are within 1% of actual attenuation for soft tissue and bone and within 3% for lung from 50 keV to 15 MeV (CIRS, 2019). The Thorax dimensions are 23 cm × 32 cm and is composed by 25 tissue equivalent slices. In each slice the most radiosensitive organs are delineated and for each organ a given number of holes for detectors placement are available (see Fig. 2). The organs considered in this work, as well as the number of slices material involved in dose assessment are reported in Table 2.

2.2. Dose assessment with TLDs measurements

In order to measure absorbed dose in the physical phantom, LiF:Mg,Cu,P (TLD-100H) Harshaw EXT-RAD TLDs were used, with a circular shape and 5 mm of diameter. The LiF detectors family presents

Table 2

Organs considered for dose assessment. For each organ, the number of TLDs used for calculations is reported. In the first column, the mass-energy attenuation coefficient at 65 KeV and for each material is reported.

Organ	μ_{en}/ρ (cm ² /g) @65 keV	N ^o of slices	N ^o of TLDs
Thyroid	0.0339	4	4
Esophagus	0.0299	3	3
Heart	0.0312	2	2
Left lung	0.0312	5	18
Right lung	0.0312	5	19
Liver	0.0308	6	32
Stomach	0.0299	3	7

wide applicability due to their advantages namely the low effective number ($Z_{eff} = 8.14$) and the different shapes, sizes and sensitivity (McKeever, 1985).

The TLDs were reset on the previous day of the irradiation and the readouts were performed on the day after using a Harshaw 6600 reader. A specific pre-defined time-temperature profile was considered to avoid any contribution of the non-dosimetric peaks (Pereira et al., 2016). To control the background, transit detectors were used, however, the background subtraction was not performed due to their negligible value. The measurement system was previously calibrated in terms of air kerma using an ISO Narrow 80 spectrum (N80) at Laboratório de Metrologia das Radiações Ionizantes, Portugal (Campus Tecnológico e Nuclear, 2019). The Kerma in air (K_{air}) was obtained according to equation (1), where the Raw Data (RD) is multiplied by the Efficiency Correction Coefficient (Ecc) of each detector, the corrections factor of reader stability ($f(Q)$), fading effect $f(fad)$, energy and angular dependence ($f(E)$ and $f(\alpha)$), and divided by the Reader Calibration Factor (RCF),

$$K_{air} = \frac{RD \times Ecc}{RCF} \cdot f(Q) \cdot f(fad) \cdot f(E) \cdot f(\alpha) \quad (1)$$

The $f(Q)$ factor was considered to be equal to one once the quality control did not present any variation. The correction due to the fading effect was not taken into account because the range between the reset and the readout was negligible. The factors of energy and angular dependence were also considered equal to one because the irradiation conditions were well known.

Under the condition of electronic equilibrium, the final dose value assessed in each position of the phantom was obtained according to the following equation:

$$D_{organ} = K_{air} \cdot \frac{(\mu_{en}/\rho)_{tissue}}{(\mu_{en}/\rho)_{air}} \quad (2)$$

where K_{air} is the Kerma in air previously defined, and $\frac{(\mu_{en}/\rho)_{tissue}}{(\mu_{en}/\rho)_{air}}$ is the ratio of mass energy absorption coefficient of tissue to air at an average energy of the spectrum of 65 keV. Web-based NIST database was used for mass energy-absorption coefficients in air and tissues reported in Table 2 (NIST.gov, 2019). The final uncertainty of the measurement (about 17%) is a combined uncertainty considering the contribution

from the detector efficiency, the stability of the quality control correction factor and the reader calibration factor.

2.3. Cancer risk incidence assessment

Radiation-related cancer risks resulting from exposures to low doses of IR are small and difficult to detect relative to the random fluctuations in baseline cancer rates or background risk and confounding factors unrelated to radiation exposure (e.g. smoking). Very large groups of individuals would have to be followed during long periods of time to ensure that sufficiently accurate estimates of cancer risk associated with low dose exposures (Martin et al., 2009; Wakeford, 2012). Alternatively, cancer risk models offer a statistical description of how the probability of cancer induction by radiation varies with the dose received by specific organs. These models take into consideration important risk modifying factors, such as gender, age at exposure and time since exposure. The cancer risk models are mostly based on the epidemiologic findings from the Life Span Studies (LSS) of the A-bomb survivors, combined with radiobiological knowledge gained from experimental studies (Wakeford, 2012; BEIR VII report, 2006).

In this study, the age-dependent and gender-specific Lifetime Attributable Risk (LAR) estimates of radiation induced cancer incidence and mortality published on the BEIR VII – Phase II Report, were used together with mean organ doses calculated for the thorax CBCT examination, in order to estimate the risk for incidence for solid cancer sites (BEIR VII report, 2006):

$$R_T = D_T \cdot 0.1 \times K^{a,g} \quad (3)$$

Where R_T is the number of cancer cases or deaths per 100 000 exposed persons for a specific organ or tissue T; D_T is the organ dose in Gy, and $K^{a,g}$ is the age (a) and gender-specific (g) LAR coefficient (cases or death per 100 000 exposed to 0.1 Gy) for organ or tissue T. The coefficients $K^{a,g}$ are tabulated in the BEIR VII report for males and females at discrete ages from 0 to 80 years, in 5 years step. The LAR coefficients for each age were determined by linear interpolation of the tabulated data.

3. Results and discussion

3.1. Organ dose assessment

Table 2 shows the numbers of TLDs in each organ considered. The absorbed dose was calculated, through formula (2) previously described, as an average of the punctual dose distribution in the organ. The TLDs were exposed to a typical Thorax protocol that is routinely used in clinical practice. The left lung was chosen as target organ (isocenter), with a source-to-isocenter distance (SID) of 100 cm. The absorbed doses calculated are reported in Table 3, both for only 1 session and for a hypothetical treatment plan composed by 24 sessions for lung cancer. Among the organs considered, the thyroid received the minimum dose (0.66 mGy), whereas the heart received the maximum one (5.84 mGy). The target organ (left lung) received a dose of

Table 3

Dose values measured with TLDs, both for only 1 session and for a cumulative hypothetical lung cancer treatment of 24 sessions.

Organ	Average Dose per session (mGy)	Total dose for hypothetical 24 sessions plan (mGy)
Thyroid	0.66 ± 0.11	15.84
Esophagus	3.56 ± 0.60	85.44
Heart	5.84 ± 0.99	140.16
Left lung	4.90 ± 0.83	117.36
Right lung	3.07 ± 0.52	73.68
Liver	3.47 ± 0.59	83.28
Stomach	3.32 ± 0.56	79.68

4.89 mGy, whereas the surrounding healthy tissues had an average dose of about 4.8 mGy. According to the work of P. Alaei et al. (Alaei and Spezi, 2015) about dose assessment (experimental and computational) with CBCT scan protocols, only one work (Kan et al., 2008) is comparable with the present work, being also performed with anthropomorphic physical phantom irradiated with a thorax protocol, TLDs and the dose values are discriminated per organ.

In M. Kan et al. (2008) work the organ doses present values that are about twice compared to those assessed in the present study. Differences could be partially explained by the different TLDs used, anthropomorphic phantom and by some technological difference in the kV-imaging system used (Sun et al., 2017). In M. Kan et al. no specification of how the X-ray spectrum was generated is supplied, whereas the specification of the system used in this work is clearly known (uncertainties on the use or not of titanium beam-hardening filters could generate dose differences). In addition, if considering the computational study performed with patient CT images as input data for MC calculations, there is a good agreement with results of this work for the studied organs (Nelson and Ding, 2014). The present study strengthens the idea that the kV-image dose in IGRT is a strongly variable of several factors, such as protocol parameters, detector used, detailed knowledge of the X-ray energy spectrum distribution among others. As further remarked by the AAPM Task Group 180, the exact knowledge of the kV-image dose is very important if considering that 5% of the therapeutic target dose is the threshold beyond which imaging dose should be accounted for treatment planning processes (Ding et al., 2018). While for a single treatment session, the imaging dose is quite lower than this 5% threshold, considering the cumulative dose for an entire treatment, the dose value could be considerable. For this reason in the results of Table 3, a hypothetical total cumulative dose estimation was also added for comparison. Considering a hypothetical lung cancer treatment planning of 24 sessions, the imaging dose could reach a value of 117 mGy in the target organ, and an average of 80 mGy in the surrounding healthy tissues (excluding thyroid). These values can significantly vary, depending on the number of planned sessions and, in a daily practice, if the kV-images are acquired in a pre and post irradiation stage (Sun et al., 2017).

The estimation of the kV-imaging dose in IGRT is not an easy task, since there is no automatic tools integrated in the imaging systems able to make an estimation of the organ doses involved in the target volume. The only dosimetric output involving the CBCT dose is the automatic estimation of the CTDI_w that however presents a series of concerns and doubts (especially in CBCT dosimetry) because is not fully able to take into account the specific absorbed doses in organs (Ding et al., 2018; Abuhaimeed et al., 2015; Martin et al., 2016). Certainly computational calculations are of paramount importance in order to estimate the organ doses in medical procedures such as CBCT. However, in order to define a dosimetric formalism for these applications, the computational calculations should be always followed by extended campaigns of measurements in a more systematic way (different protocols, different systems, etc.).

3.2. Cancer incidence assessment

Considering both Thorax CBCT scanning protocol, the risk estimates of cancer incidence was determined (according to the BEIR-VII cancer risk models) because of the cumulative CBCT imaging doses after a complete course of radiotherapy treatment. The CBCT scanning protocol evaluated in this work are only applicable for adult patients, whereby the risk estimation was determined for ages between 18–80 years. The risk was estimated for the target organ and for the radio-sensitive surrounding organs at risk that received higher CBCT doses. Fig. 3 (left) shows the cancer incidence coefficients as reported by the BEIR VII report for lung, liver, stomach and thyroid, whereas Fig. 3 (right) shows the cancer incidence values being interpolated in the 18–80 years interval (step by one year) and weighted according the

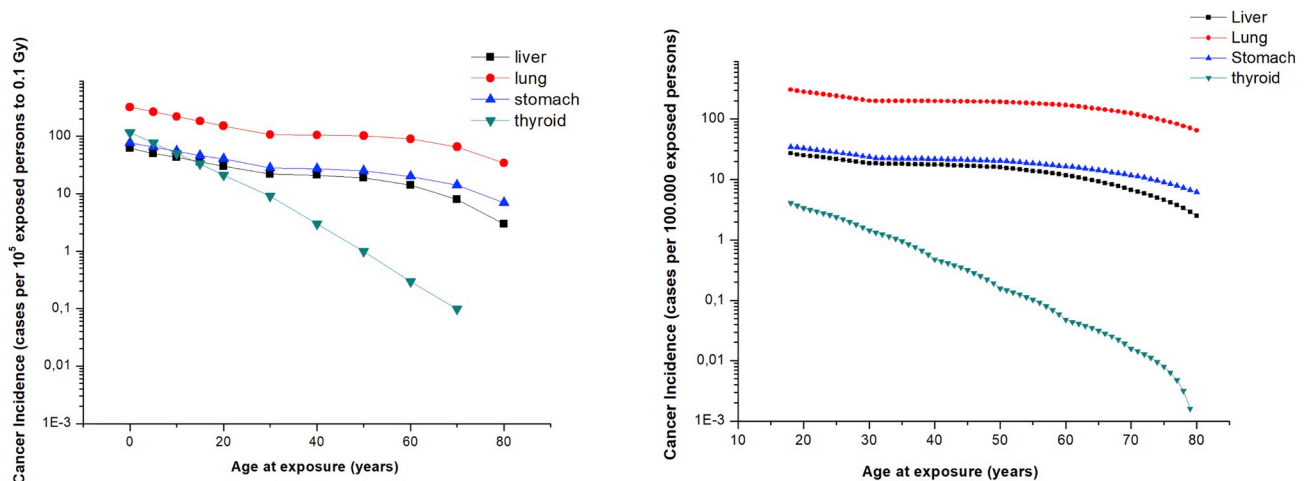


Fig. 3. Left) Cancer incidence values as extracted by the BEIR VII report in the 0–80 years age interval; Right) Cancer incidence values weighted with the absorbed doses assessed with TLDs and reported in Table 3.

dose values calculated in Table 3. The cases of lung cancer incidence span from a minimum of $65/10^5$ (at 80-year age) to a maximum of $309/10^5$ exposed individuals (at 18-year age). The cases of stomach and liver cancer incidence span from a minimum of about $6/10^5$ exposed individuals (at 80-year age) to a maximum of about $30/10^5$ (at 18-year age). Finally, the cases of thyroid cancer incidence span from a minimum of $0.003/10^5$ exposed individuals (at 80-year age) to a maximum of $4/10^5$ (at 18-year age).

Analyzing these data risk, it is safe to say that, in this specific case, in average the probability to develop lung cancer is mainly distributed in the 40–60 year age interval for adult male. For this reason, when looking at the lung cancer risk, some average value should be considered as indicative in this age interval. As shown in Fig. 3 the probability to develop cancer after such type of exposure is quite low, even for the target organ (Donovan et al., 2012). In addition, cancer risk models present several sources of uncertainties that should be taken into account (i.e. poor knowledge of the shape of the dose response curve below 100 mGy, presence of confounding factors such as smoking, radiation dosimetry, etc. (Baptista, 2019)). However, given the cumulative dose effects, these cancer risk trend can be useful especially in decision-making concerning organ tumor (both target and healthy tissues) treatment planning.

4. Conclusions

The two main aims of this work were to assess organ doses and risk of cancer development for a typical Thorax scan protocol clinically used in IGRT. There are relatively few works about organ dose estimation with physical anthropomorphic phantoms exposed to a Thorax IGRT protocol. In addition, to best of our knowledge, this is the first dosimetric study performed with physical phantom in a CBCT Varian Edge system. The results of this study confirm a sort of scatter trend about dose values, both with computational and experimental ones. Nevertheless, especially in CBCT and considering the proposed limit of 5% of the therapeutic dose, there is a need to define a better dosimetric formalism that would be able to estimate the imaging dose in the target and surrounding organs in a more precise and systematic way. For this reason, according to the authors of this work, additional and more comprehensive experimental campaigns should be performed in clinical environments for different CBCT protocols and for different equipment for several manufacturers in order to define reliable dose intervals as dose references in CBCT imaging (taking into account all the uncertainties involved) and also for the sake of accurate computational models validation.

Finally, the probability to develop cancer for this type of imaging

technique is not still deeply debated in literature and these results can be useful for better optimize the different CBCT protocol scans (both in terms of dose and image quality), especially when considering the cumulative dose for a complete treatment and generally, for the radiological protection of the patient.

CRedit authorship contribution statement

Mariana Baptista: Conceptualization, Methodology, Investigation. **Salvatore Di Maria:** Conceptualization, Investigation, Writing - original draft, Writing - review & editing. **Sandra Vieira:** Investigation, Resources. **Joana Pereira:** Investigation, Resources. **Miguel Pereira:** Investigation, Resources. **Pedro Vaz:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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